Course Title	Digital Logic Design	**Course Code**	EE1005
Department	Department of Electrical Engineering (DEE)	**Campus**	Lahore
Knowledge Profile	Engineering Design (WK5)	**Credit Hrs.**	3+1
Knowledge Area	Computer Engineering (KA01)	**Grading Scheme**	Relative
HEC Knowledge Area	Engineering Foundation	**Applicable From**	Fall 2025
SDG	4 | Quality Education	**PBL**	1
Pre-requisite(s)			

Course Objective	To introduce the classical logic design techniques for analysis and design of combinational and sequential circuits.

No.	Assigned Program Learning Outcome (PLO)
1	Apply knowledge of mathematics, natural science, engineering fundamentals and Engineering specialization to the solution of complex engineering problems
2	Identify, formulate, conduct research literature, and analyze complex Engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences
3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

A = Assignment, Q = Quiz, M = Midterm, F=Final

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	**Describe** various number systems and perform arithmetic operations and base conversions.	M1, F	C2	1
2	**Apply** Boolean Algebra and K-map methods to optimize logic circuits	A1, Q1, M1	C3	1
3	**Design** combinational circuits using functional blocks	A2, Q2, M2, F	C6	3
4	**Analyze** the elements of sequential circuits	Q3, F	C4	2
5	**Design** different sequential circuits including counters for various applications and basic architecture of arithmetic logic unit and FPGA	A3, F	C6	3

Text Books	**Title**	Logic and Computer Design Fundamentals (5th Edition, 2015)
	Author	M. Morris Mano & Charles R. Kime
	Publisher	Prentice Hall
Reference Books	**Title**	
	Author	
	Publisher	

Week	Course Contents/Topics	Chapter*	CLO*
01	DIGITAL COMPUTERS AND INFORMATION, Digital computers and Binary Numbers, Other base numbers (base-8, base-16 etc.), Number base conversions.	1	1
02-05	COMBINATIONAL LOGIC CIRCUITS, Binary Logic and Logic Gates, Axioms and Basic Theorems of Boolean Algebra, Boolean Functions and their implementation, Canonical and Standard Forms (Minterms, Maxterms, Conversions), Minimization of Boolean functions using basic theorems and K-Maps, Don't Care States, Universal gates and implementation of Boolean functions using universal gates	2	2
05-09	COMBINATIONAL LOGIC DESIGN, Combinational Circuits, Decoders / Encoders, Multiplexers / Demultiplexer, Binary Adders (Half Adder, Full Adders, Binary Ripple Carry Adder), Binary Subtractor, Binary Adder/Subtractor, Code Conversion, Magnitude Comparator.	3	3
10-14	SEQUENTIAL CIRCUITS, Sequential Circuits (FSM), Latches, Flip Flops, Type of Flip Flops, characteristics equations and tables, State diagrams and state tables. REGISTERS AND COUNTERS, Registers, Counters, Synchronous/Asynchronous, Shift Registers, Serial Shift Registers, Loading Registers, Parallel Registers, Ripple Counters, Synchronous Binary and other Counters, Johnson's counter	4, 6	4,5
15-16	Verilog for digital circuits, simple arithmetic logic units and introduction to FPGA	Material will be provided	5

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes, Assignments	20%
Midterm (I+II)	15% + 15%
Final Exam	50%